

Skills Summary

Building Multi-Step Arguments from Givens

Use this reference while completing your worksheet or when studying.

Skill 1 — Putting the steps in a working order

Every argument has the same shape: start at a given, apply a definition to turn words into symbols, apply a postulate or theorem to get a relationship, then use properties of equality to finish.

Ordering test: a line may be written only if everything it needs is already above it. If a line needs something below it, it is in the wrong place.

Two reasons that always come as a pair: the *definition of congruent* segments or angles turns a congruence mark into an equation; the *definition* that produced the congruence comes first.

Example: Two lines cross. Given that a pair of vertical angles measure $(5x - 4)^\circ$ and $(3x + 12)^\circ$, find x .

Step 1. The given intersection lets me name the pair vertical, the vertical angles theorem then gives congruence, and the definition of congruent angles turns that into an equation — three steps in that order.

Step 2. $5x - 4 = 3x + 12$, so $2x = 16$.

$$\Rightarrow x = 8$$

Try it: Vertical angles measure $(6x + 1)^\circ$ and $(4x + 15)^\circ$. Find x .

check: $x = 8$

Skill 2 — Segment arguments

Definition of midpoint: M on $\overline{AB}$ with $\overline{AM} \cong \overline{MB}$; then $AM = MB$ and $AB = 2AM$.

Segment Addition Postulate: B between A and C gives $AB + BC = AC$.

Overlapping segments: add the shared middle stretch to both sides of a given equality, then rename each sum with the postulate. Subtracting it runs the same argument backwards.

Example: M is the midpoint of $\overline{AB}$ with $AM = 3x + 5$ cm and $MB = x + 17$ cm. Find x .

Step 1. The midpoint given produces congruent pieces, and congruent segments have equal lengths — that is the step that turns the picture into an equation I can solve.

Step 2. $3x + 5 = x + 17$, so $2x = 12$.

$$\Rightarrow x = 6$$

Try it: M is the midpoint of $\overline{AB}$ with $AM = 5x - 4$ cm and $MB = 2x + 11$ cm. Find x .

check: $x = 6$

Skill 3 — Angle arguments with two links

Linear Pair Postulate: adjacent angles whose outer sides form a line have measures adding to 180.

Vertical Angles Theorem: the opposite pair formed by two intersecting lines are congruent.

Definition of an angle bisector: the whole angle is *twice* either half.

Chaining two of these is the usual multi-step task: use the bisector to write the whole angle in terms of the marked half, then feed that into the linear-pair sum.

Example: $\overrightarrow{QS}$ bisects $\angle PQR$ with $m\angle PQS = (3x + 5)^\circ$, and $\angle PQR$ and $\angle RQT$ form a linear pair with $m\angle RQT = (4x)^\circ$. Find x .

Step 1. The bisector lets me write the whole angle as twice the marked half, and the linear pair then supplies one equation containing only x — two links, one unknown.

Step 2. $2(3x + 5) + 4x = 180$, so $6x + 10 + 4x = 180$ and $10x = 170$.

$$\Rightarrow x = 17$$

Try it: Same picture with $m\angle PQS = (2x + 10)^\circ$ and $m\angle RQT = (6x)^\circ$. Find x .

91 = x :check